Fetal biometry annotation guidelines

M. Bach Cuadra

CIBM SP CHUV-UNIL, Department of Radiology

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CIBM.

Introduction

Goal

This document establishes the **guidelines towards the standardization of the 2D biometry measurements to be done by MDs junior (supervised by a senior) or experts.** It is adapted from previous works in:

1. M. Khawam *et al.*, "Fetal Brain Biometric Measurements on 3D Super-Resolution Reconstructed T2-Weighted MRI: An Intra- and Inter-observer Agreement Study," *Front. Pediatr.*, vol. 9, p. 639746, Aug. 2021, doi: 10.3389/fped.2021.639746.

This document was originally set in the context of the ERANET MULTIFACT project for the standardization of annotations among different clinical centers (La Timone Marseille, BCNatal Barcelona, KISPI, Zurich and CHUV Lausanne) and involved researchers in that consortium.

Selected biometric measures

In sagittal view :

1. LCC, **length of the corpus callosum,** measured in the mid-sagittal plane.
2. HV, **height of the vermis,** measured in the mid-sagittal plane. Proceed the same as usually done in US: do the measurement with a line that goes parallel to brainstem.

In axial view:

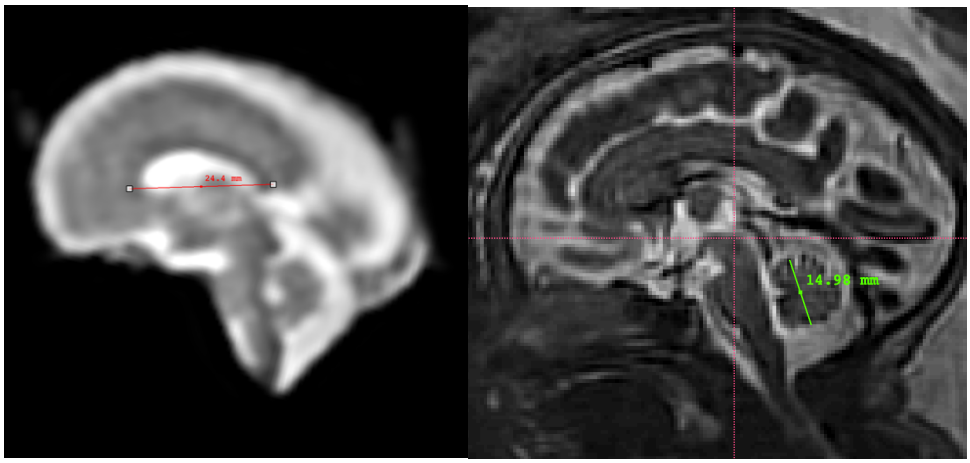
1. bBIP_ax, **brain biparietal diameter :** defined as the maximal brain diameter in the transverse plane through the atria.
2. sBIP_ax, **skull biparietal diameter:** the inner-to-inner table maximal skull diameter in the transverse plane through the atria.

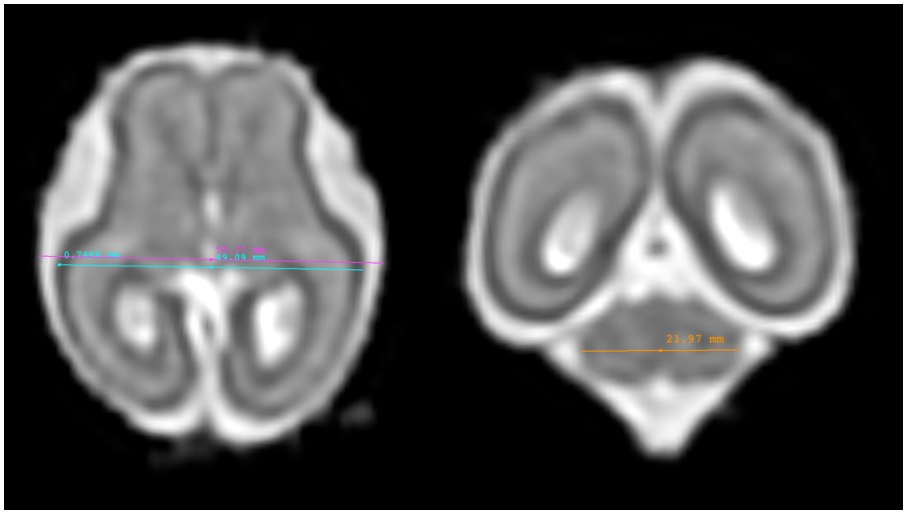
In coronal view:

3. TCD_cor, maximum **transverse cerebellar diameter**

These colors will be used please refer to page 12 for further info:

- Red : LCC
- Green : HV
- Cyan : bBIP_ax
- Magenta : sBIP_ax
- Orange : TCD_cor





How to use of ITK-SNAP

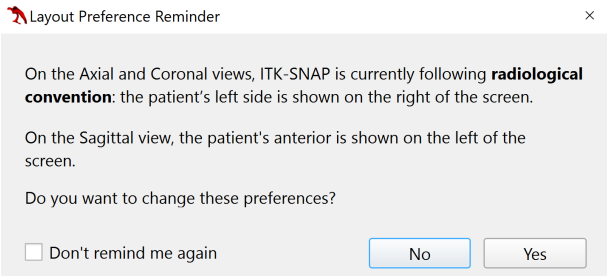
This tutorial is based on ITK-SNAP 4.x Team (2020 - present), using a Mac OS:

- http://www.itksnap.org/pmwiki/pmwiki.php

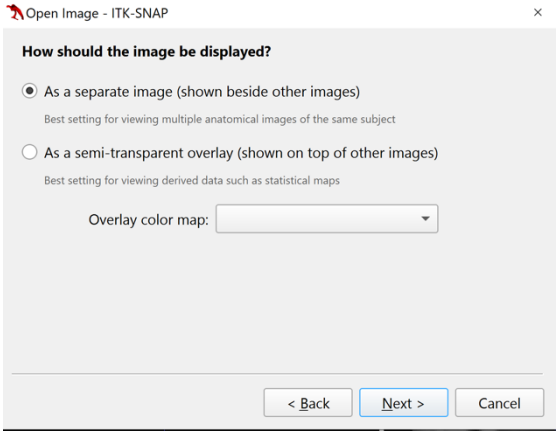
If need to re-orient the images

Warning: Reorientation of the images will cause blurring due to interpolation, so **if you can avoid this step would be preferable.**

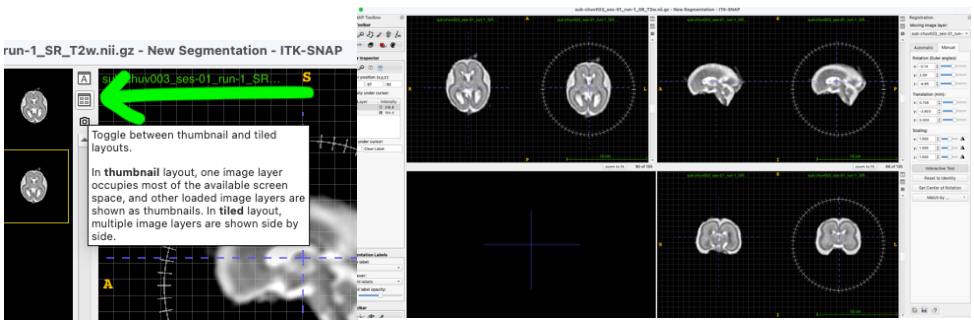
- Load the image as main image and check the main orientation preference



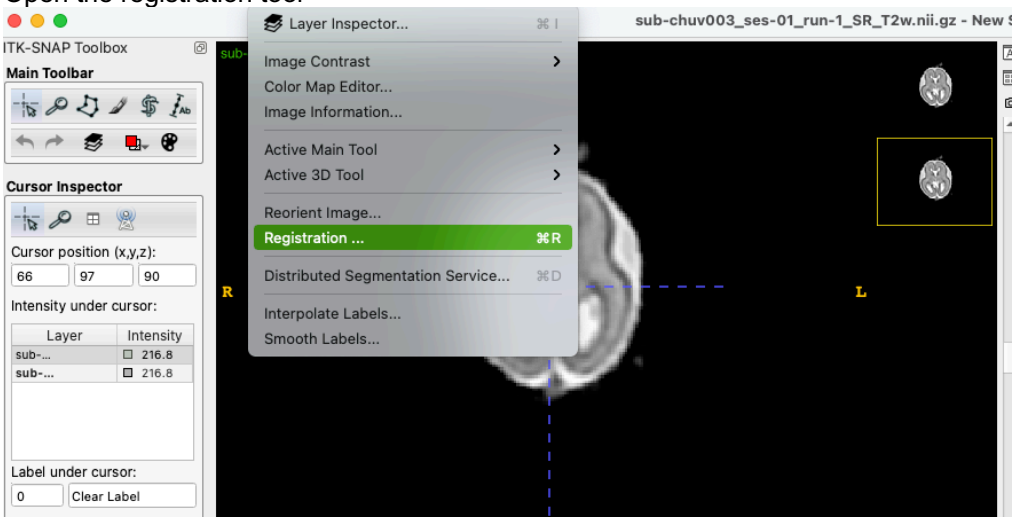
- Left click at the center of the MRI to locate the center of the blue cross at the center of the brain, which will help to reorient the image later
- Load the same image again as "Additional Image" and display it as a separate image



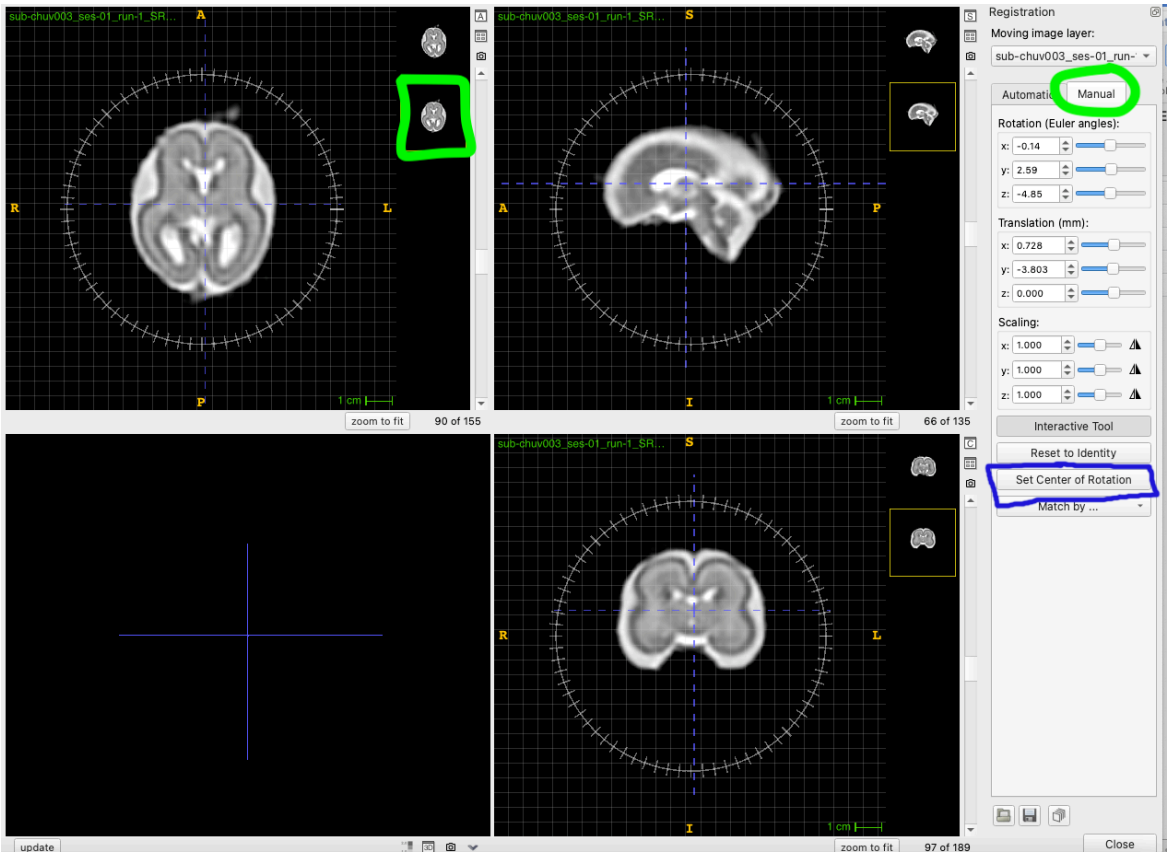
- You can choose between two types of visualization by clicking on the thubnail button: if you prefer you can visualize the two image simultaneously by clicking tiled layouts:



- Open the registration tool

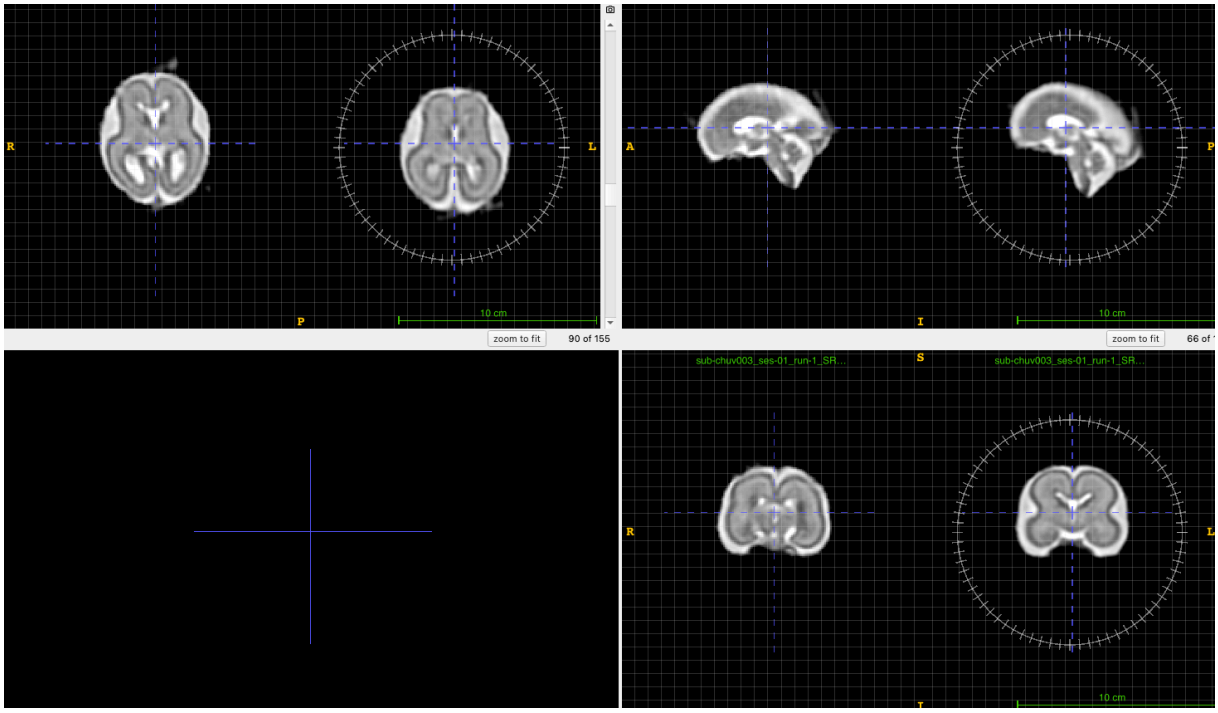


- Ensure you are in Manual registration mode (check on the green squares here below), and click on the button 'Set Center of Rotation' (blue square below) if the brain is not properly located within the circle:



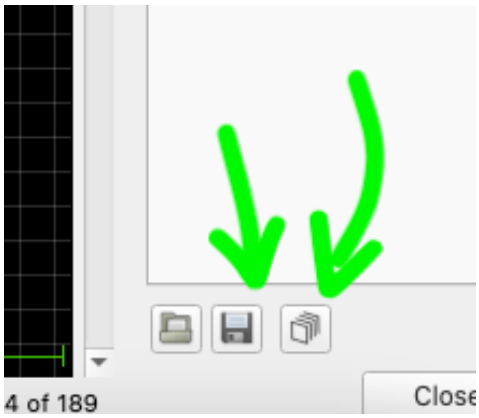
- You can rotate / translate the image either by using the toggle bars on the right side, or in a very intuitive way using your mouse over the image:
    - Left click on the circle will let you rotate in any plane

- Left click and move the image will translate it

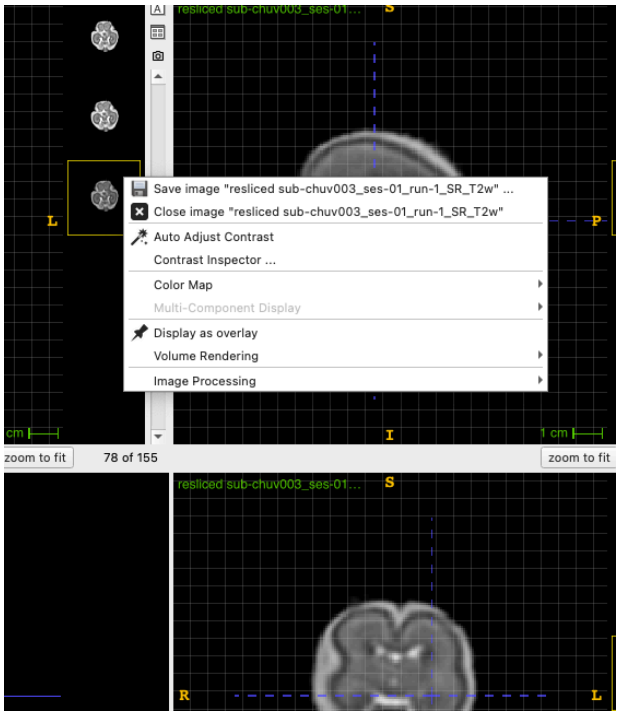


- Once the preferred alignment is obtained, you can save both the transformation and the realigned image by clicking at the bottom the icons disc (save transformation) and multiple slices (reoriented image with linear interpolation).

This is recommended as otherwise we cannot ensure the reproducibility of the measurements to have the exact reorientation retrieved.



- Save the transformation as *name-of-the-image_reorient.txt*, save the resliced image (from the viewing) as *name-of-the-image_reorient.nii*

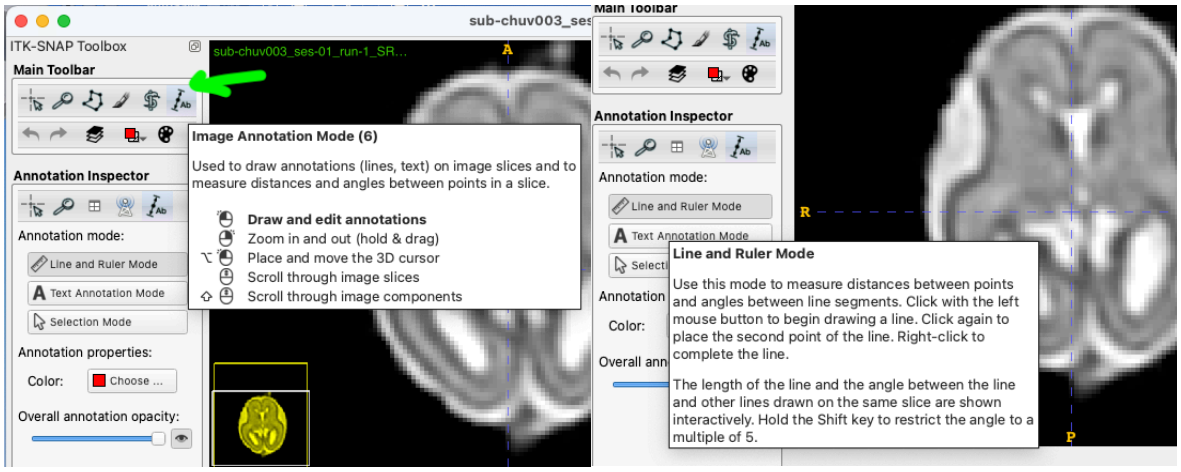


- Close itksnap and reopen the resliced image
- You can now navigate and proceed to the annotations as described below.

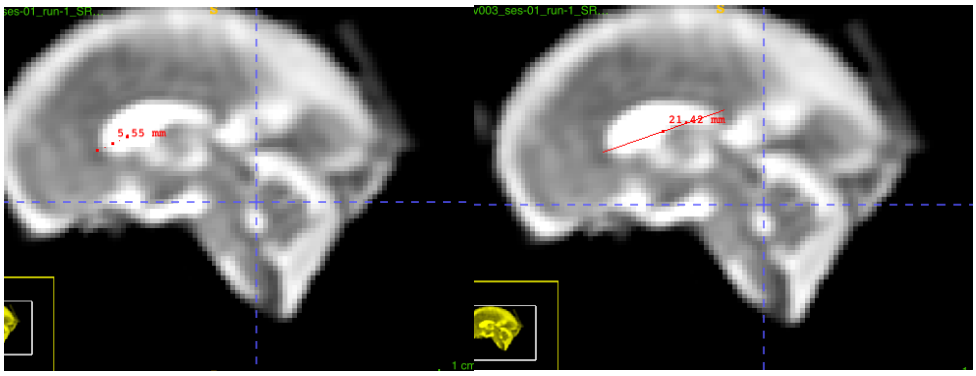
Distance measurements

We will use the **Line and Ruler mode** of the **Annotation Inspector:**

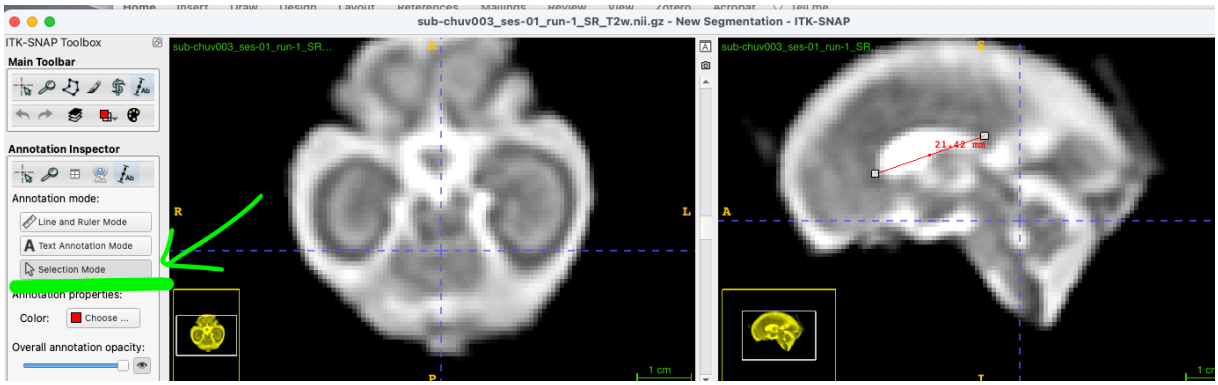
- Click on the Image Annotation Mode from the main toolbar (icon of a measure bar with Ab letters)
- Get familiar with the mouse button usage to navigate withing the image
- Tip: Key 'x' make disappear/appear toggle the blue cursor in the screen
- Line and Ruler Mode must be selected:



- For each measurement, **one specific color needs to be selected**. E.g. we start with the default color (red) for measuring LCC
- Navigate through the sagittal plane
- Click on the left mouse button to start the measure (left panel here below), unclick to end the measure (right panel here below)



- If you want to edit/remove your measurement: Click on the **selection mode (icon white arrow)**
- You can then either adapt your measure or remove it with « backspace » key from your keyboard

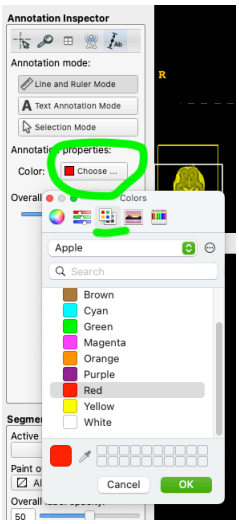


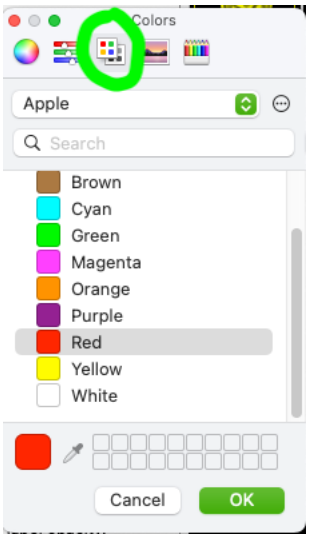
- Come back to Line and ruler Mode (ruler icon)

- Before proceeding to next measure you can **take a screenshot** and save it into the subject folder
    - With Mac : shift+cmd+3
    - With Windows: win+shift+s

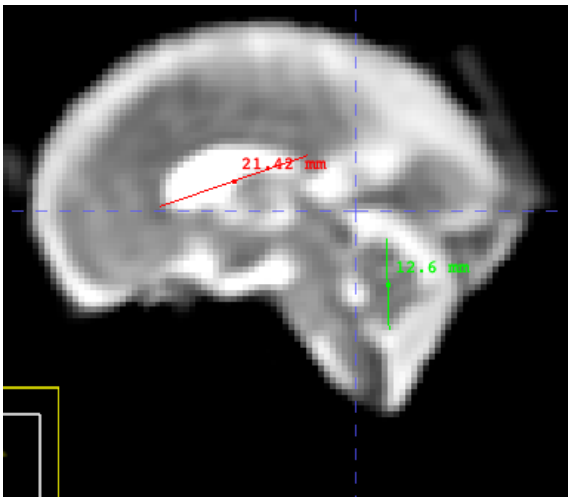
- Now before proceeding to next measure, **select the next color**

- Click on Chose Color, ensure you have the good type of "display" colors (list of pre-defined colors)

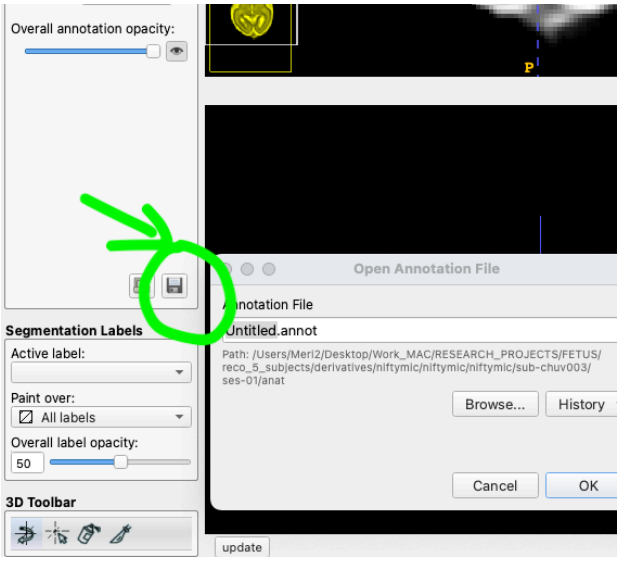




- Select the new color, Green for *vermis height* selection
- Click ok
- Do the measurement as before, right click to start the measure, drop click to end the measure



- Please ensure the following colors are used for the biometry :
    - Red : LCC
    - Green : HV
    - Cyan : bBIP_ax
    - Magenta : sBIP_ax
    - Orange : TCD_cor
- When all the measurements are finished <u>SAVE THE ANNOTATIONS !!!!!</u>
- Click on the disk Icon at the left :



- Select as file name the name-of-the-image_*yourInitials.annot*, eg. : chuv-003_MBC
- Ensure the annotation file is stored in the correct/desired directory